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A rotatable cathode with tunable steric hindrance for high-performance aluminum organic batteries†

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Rechargeable aluminum batteries (RABs) with uniquely high energy-to-price ratios are promising energy storage systems. However, owing to the large-size active ions (e.g. AlCl_4^- and $[\text{AlCl}_2(\text{urea})_2]^+$) in state-of-the-art ionic-liquid RABs, the discharge–charge process usually encounters severe steric hindrance on electrodes, resulting in insufficient active-site utilization with limited practical capacity, and sluggish ion-diffusion kinetics with inferior rate-performance. Herein, to overcome these intrinsic confinements from the large-size active ions, we proposed, for the first time, a structure-rotatable cathode (quinone-based calix[4]quinone (C4Q)) for RABs with highly reduced steric hindrance, which is further verified *via* density functional theory (DFT) simulations. The newly applied C4Q cathode achieves one of the highest energy densities (480 W h kg^{-1} at 0.1 A g^{-1}), enhanced rate capacity (81 mA h g^{-1} at 1.0 A g^{-1}), and long-term stability (102 mA h g^{-1} at 0.2 A g^{-1} after 500 cycles) in RABs. Furthermore, the reaction mechanism of this rotatable cathode is revealed in detail *via* a series of characterization studies and corresponding DFT simulations. Overall, this new family of rotational organic materials will provide a new direction for promising high-performance RABs.

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Introduction

Rechargeable aluminum batteries (RABs), especially non-aqueous ionic-liquid (IL) RABs, have attracted ever-increasing attention as promising energy storage systems due to their theoretically high capacity ($2980 \text{ mA h g}^{-1}/8040 \text{ mA h cm}^{-3}$), low cost, and high natural abundance.^{1–3} Thus, considerable effort has been invested in RABs during the last few decades, especially the cathode sides,⁴ such as metal oxides,^{5,6} metal sulfides/selenides,^{7–9} Prussian blue analogs,¹⁰ MXenes,¹¹ and metal–organic framework materials.^{12,13} Due to the high-charge-density of Al^{3+} (364 C mm^{-3} , more than six times higher than that of Li^+), these traditionally rigid-structure rocking-chair-type cathodes usually have high electrostatic interaction with Al^{3+} ,^{14–16} resulting in serious crystal strain and grievous structure pulverization with inferior cycling stability.^{17–19} In comparison, the novel monovalent species such as AlCl_4^- and $[\text{AlCl}_2(\text{urea})_2]^+$ allow facile reversible intercalation in state-of-the-art ionic-liquid RABs. Since being reported in 2015,^{20,21} varieties of graphite-based cathodes have been attempted in RABs.^{22,23} However, their inherently narrow interlayer space and

limited active-sites still hinder practical development with an unsatisfactory capacity less than 150 mA h g^{-1} .²⁴

Notably, organic materials, with their advantages of a designable molecular structure, environmental friendliness, sustainability and intrinsically appropriate intermolecular interactions, are especially suitable for large-size monovalent aluminum carrier ions.^{25–28} Thus, considerable effort has been focused on the development of organic-based cathodes, including carbonyl group-based materials,²⁹ pyrazine/imine group-based materials,^{30,31} cyano group-based materials,³² and metal–organic frameworks (MOFs).^{17,33} Amongst them, quinone-based materials have attracted much attention due to their multiple redox feeding sites and high oxidation/reduction potential, regardless of *para* or *ortho* quinone. For example, Yoo and co-workers reported the phenanthrenequinone (PQ) triangle and tetradiketone,^{34,35} revealing that the storage mechanism of Al is to share two carbonyls to stabilize the structure. Multiple carbonyls are beneficial for the specific capacity of traditional organic materials, while the closely neighboring carbonyl groups usually limit the utilization efficiency of the active sites, in which the two or three adjacent active sites can only, in combination with one large Al-based active ion with hindered capacity. However, multiple carbonyl materials like C4Q are widely applied in aqueous aluminum and zinc batteries and achieve a good performance.^{36,37} The active ions in the IL system, however, are larger than the active ions in the aqueous system. Due to large-size active ions, such as AlCl_4^- and $[\text{AlCl}_2(\text{urea})_2]^+$, traditional organic cathodes still

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encounter insufficient space, serious steric resistance between active ions, poor active-site utilization with low energy density, and inferior diffusion kinetics. As a result, it is crucial to investigate material electrodes with rotatable structures, multiple active sites, and low steric hindrance for the development of organic aluminum batteries. C4Q can rotate in space to some extent, which is more favorable for the large active ions in the IL-based system to play a more significant role, yet been reported in IL-based RABs.

Herein, we propose, for the first time, a rotatable structure of multiple carbonyl quinone (calix[4]quinone (C4Q)) as the cathode material for RABs. The newly applied C4Q has the intrinsic advantage to reduce steric resistance and adjust the obstruction from ever-increasing reactive-ions. Benefiting from the intrinsically multiple redox sites and flexible structure, C4Q can accommodate quantities of large-size ions with less steric hindrance, further verified *via* density functional theory (DFT). The newly developed C4Q cathode achieves one of the best energy densities of 480 W h kg^{-1} and robust cycling stability (102 mA h g^{-1} at 0.2 A g^{-1} after 500 cycles) in RABs. In addition, the corresponding DFT simulation and a series of process characterization studies further reveal the reaction mechanism. Consequently, the high-performance rotatable organic-based

cathode shows bright prospects for electrode development in RABs.

Results and discussion

Explicitly, higher spatial resistance is seen in materials with rigid structures; since one of the carbonyl ($\text{C}=\text{O}$) groups reacted with the large Al-based active ions ($[\text{AlCl}_2(\text{urea})_2]^+$), the remaining active-site is hard to be effectively utilized due to high steric resistance (Fig. 1a).³⁸ The two adjacent carbonyl groups on the benzene ring are favorable for specific capacity. However, during the reaction, since the positions of the carbonyl group on the benzene ring are fixed, after the reaction of one of the carbonyl groups, the remaining space may be insufficient to accommodate a second reactive ion to react with another carbonyl group. Therefore, high steric hindrance results in unsatisfactory performance. Interestingly, the rotatable C4Q consists of four benzoquinone units and four “ $-\text{CH}_2-$ ” with a three-dimensional chair structure (Scheme S1†),³⁹ and the “ $-\text{CH}_2-$ ” group in the structure acts as a hinge to tune the space steric hindrance, which can enhance the active site utilization and accommodate large-size active ions in RAB systems. Fig. 1b shows the charge/discharge process of the C4Q

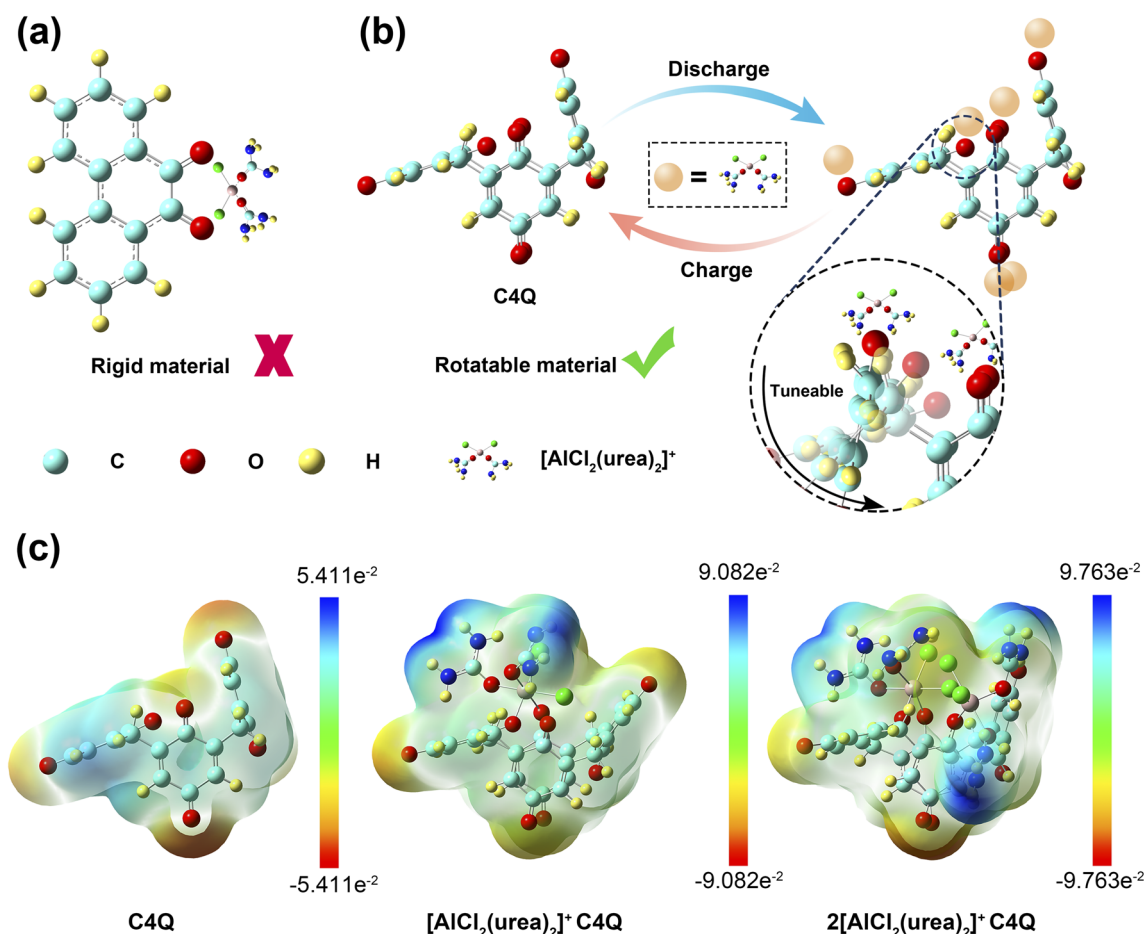


Fig. 1 Reaction process of (a) traditionally rigid organic materials and (b) C4Q. (c) Molecular electrostatic potential (MESP) process of C4Q coordination with $[\text{AlCl}_2(\text{urea})_2]^+$ and $2[\text{AlCl}_2(\text{urea})_2]^+$.

cathode with the oxidation/reduction of the C=O group and $[\text{AlCl}_2(\text{urea})_2]^+$ accompanied by multiple electronic reactions.⁴⁰ Upon reduction, the electrolyte ion $[\text{AlCl}_2(\text{urea})_2]^+$ approaches C4Q and bonds to the active site. Moreover, the molecular electrostatic potential (MESP) is obtained to demonstrate the bonding interaction between $[\text{AlCl}_2(\text{urea})_2]^+$ and C4Q (Fig. 1c). The MESP distributions of C4Q indicate multiple spots with a higher negative value around the O atoms, which are viewed as possible active sites with high chemical affinity to absorb $[\text{AlCl}_2(\text{urea})_2]^+$. This result revealed that the positively charged $[\text{AlCl}_2(\text{urea})_2]^+$ was more likely to bind with the negatively charged C=O groups in the C4Q molecule due to the significant intermolecular forces. As one $[\text{AlCl}_2(\text{urea})_2]^+$ active ion is presented, the optimal geometrical arrangement ($[\text{AlCl}_2(\text{urea})_2]^+$ C4Q) predicts a region with high electronegativity concentration near the C=O groups, which might easily attract other $[\text{AlCl}_2(\text{urea})_2]^+$ ions to further coordinate with C4Q. When coordinated with $2[\text{AlCl}_2(\text{urea})_2]^+$, this electronegativity improvement may be sustained ($2[\text{AlCl}_2(\text{urea})_2]^+$ -C4Q). Besides, as the structure changes from C4Q to $[\text{AlCl}_2(\text{urea})_2]^+$ -C4Q to $2[\text{AlCl}_2(\text{urea})_2]^+$ -C4Q, the benzene ring's angle about C4Q as a whole is plainly altered. Because when $[\text{AlCl}_2(\text{urea})_2]^+$ is bound with adjacent C=O groups, C4Q would rotate by a specific angle through the tunable “-CH₂-” to reduce the steric hindrance and promote the binding of other $[\text{AlCl}_2(\text{urea})_2]^+$ ions. Thus, the rotatable structure of C4Q can not only improve the utilization rate of the carbonyl groups but may also adjust to the role of large-size active ion $[\text{AlCl}_2(\text{urea})_2]^+$ during the charge/discharge process.

According to the ¹H nuclear magnetic resonance (NMR) spectrum, two typical peaks of C4Q are exhibited at 3.47 and 6.71 ppm, belonging to the “-CH₂-” and “-CHC-” groups (Fig. S1a†). And the ¹³C spectra show the peaks of 29.49, 134.16, 146.77, 185.77, and 187.70 ppm belonging to the C4Q sample (Fig. S1b†). The peak concentrations correspond to those previously reported in the scientific literature.⁴¹ Meanwhile, the electrospray ionization mass spectrometry (ESI-MS) spectrum describes the characteristic peaks of C4Q consistent with previous literature reports (Fig. S2†). Besides, in the X-ray diffraction (XRD) pattern displayed, C4Q presents strong diffraction peaks consistent with literature reports (Fig. S3†).⁴² Moreover, the molecular vibrational signals of C4Q were analyzed by Fourier transform infrared (FTIR) and Raman spectroscopy. The FTIR spectrum shows that the peak at 1295 cm⁻¹ belongs to C-H bending, and the peaks at 1612 cm⁻¹ and 1657 cm⁻¹ are ascribed to the stretching vibrations of C=O bonds in C4Q (Fig. S4†).⁴³ The Raman spectrum reveals the dominant peaks located at 1630 cm⁻¹ and 1668 cm⁻¹ corresponding to the C=O group (Fig. S5†); the peak at 1190 cm⁻¹ is associated with the vibration mode of the “C-H” group. Furthermore, the X-ray photoelectron spectroscopy (XPS) spectrum verifies the different chemical environments in C4Q, including the C=C, C-C, and C=O groups and π-π* band (Fig. S6†).⁴⁴ The obtained results demonstrated that C4Q was successfully prepared.

Additionally, the thermostability and morphology of C4Q powder are also investigated by thermogravimetric (TG) analysis and scanning electron microscopy (SEM), respectively. As

shown in Fig. S7,† the TG curve includes two stages with a slight weight loss of water evaporation (under 100 °C) and decomposition process (above 260 °C). The high degradation temperature suggests a fine thermostability ascribed to the structure of C4Q. Besides, the SEM image of the C4Q powder exhibits a micron-sized granular appearance (Fig. S8†). The transmission electron microscope (TEM) results exhibit a similar granular appearance, and the elemental mapping reveals that the C and O elements exhibit uniform distribution (Fig. S9†). It can be speculated that the bulk, rotatable and large-cavity structures of C4Q are more conducive to the role of multivalent ions (such as trivalent Al³⁺).

To minimize the volume expansion and enhance the electrical conductivity of the C4Q cathode for the Al/C4Q organic battery, multiple-branched Ketjen black carbon (KB-C) was used as the conductive agent to retain the comprehensive contact between C4Q and KB-C. Subsequently, the electrochemical performance of C4Q as an aluminum battery cathode is first evaluated with AlCl₃/urea as the electrolyte (Fig. 2). After the battery had stabilized, cyclic voltammetry (CV) was carried out. Fig. 2a shows the CV curves of the C4Q cathode with a voltage window from 0.9 V to 2.0 V at a scan rate of 0.4 mV s⁻¹. Two pairs of redox peaks can be seen on the sample's CV curves at 1.6/1.9 and 1.5/1.8 V vs. Al/Al³⁺, which are likely the result of interactions between the large-size active ions ($[\text{AlCl}_2(\text{urea})_2]^+$). The narrow redox peaks and small peak separations indicate the small polarization phenomenon of the C4Q cathode. Moreover, the curves were almost identical without much change from the three-circle scan. In comparison, the CV curves of the Mo foil and AlCl₃/urea (1.3 : 1, M/M) display negligible redox peaks (Fig. S10a and b†), further suggesting that the two pairs of oxidation/reduction peaks originate from the C4Q cathode. In addition, the galvanostatic discharge/charge curves of the C4Q cathode are presented in Fig. 2b and S11.† The charge/discharge plateaus are in accordance with the peaks in CV tests, corresponding to the multiple oxidation/reduction behaviors of the C4Q electrode. Specifically, the C4Q cathode possesses an excellent initial discharge specific capacity of up to 300 mA h g⁻¹ at a current density of 0.1 A g⁻¹. Additionally, the specific capacity of KB-C is less than 30 mA h g⁻¹ with negligible capacity contribution (Fig. S12†). Meanwhile, a classical non-rotatable cathode (PQ) was used in the same system with C4Q to investigate the utilization of active sites and the practically high specific capacity. The results demonstrate that the practical, specific capacity of the non-rotatable PQ is substantially lower than the theoretical specific capacity, which would be attributed to the low active site utilization caused by the large steric hindrance (Fig. S13†). Compared to the electrochemical performance of non-rotatable materials, the active site utilization is significantly improved (Table S1†). Thus, the high specific capacity of the C4Q cathode probably contributed to the high utilization of multiple redox sites in rotatable C4Q. Besides, the C4Q cathode also exhibits excellent rate performance, which reaches 300, 261, 174, 113, and 81 mA h g⁻¹ at current densities of 0.1, 0.2, 0.3, 0.5, and 1 A g⁻¹, respectively (Fig. S14†). What is more, the C4Q cathode's high specific

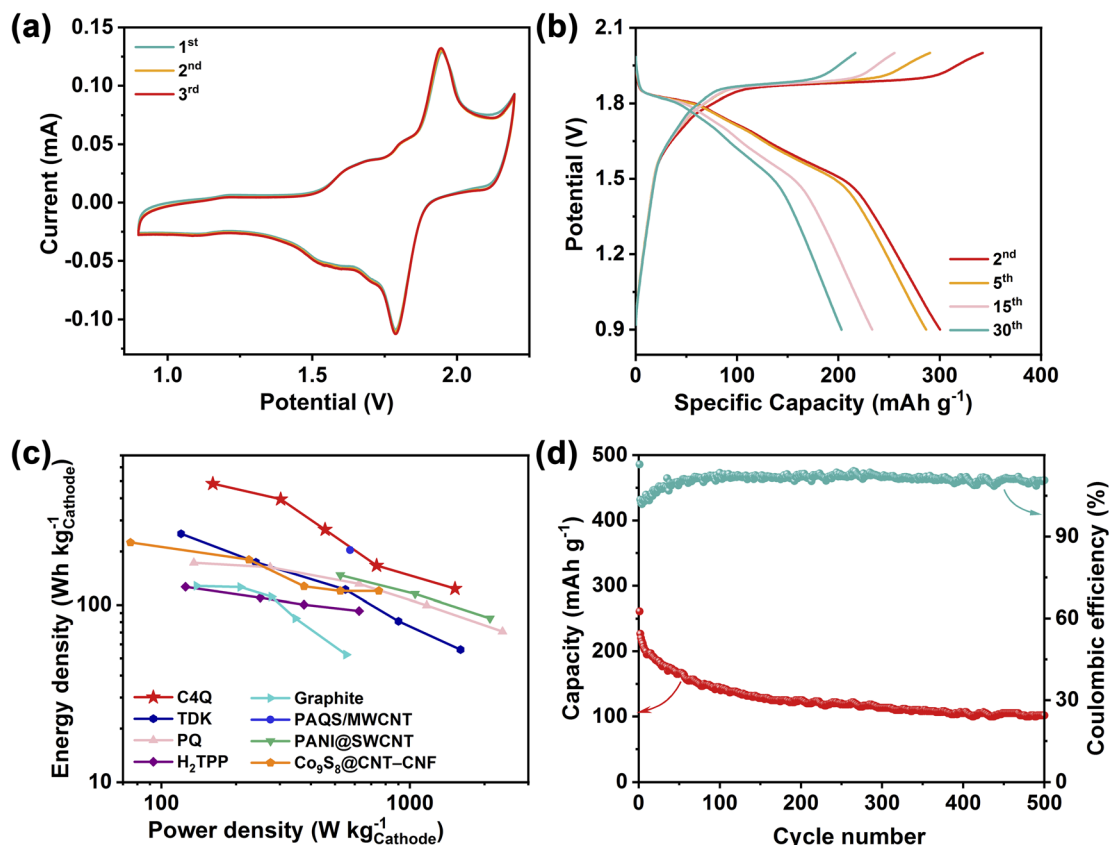


Fig. 2 Electrochemical performance of C4Q in RABs. (a) Cyclic voltammograms profiles; (b) galvanostatic discharge/charge profiles of C4Q electrodes at a current density of 0.1 A g^{-1} ; (c) Ragone plots of C4Q and other representative cathode materials of RABs; (d) long-term stability of cycling C4Q at a current density of 0.2 A g^{-1} .

capacity and redox potential result in a high energy density of up to 480 Wh kg^{-1} (Fig. 2c). Note that the energy density and power density shown in Fig. 2c are based on the cathode. To our knowledge, the C4Q cathode possesses one of the best energy densities among previously reported graphite materials, metal sulfides, and small organic molecules for RABs (Table S2, Fig. S15[†]).^{20,34,35,45–48} Besides, a battery with C4Q as the cathode material has superior storage stability and reduced self-discharge impact (Fig. S16[†]). Additionally, the open circuit voltage (OCV) at a constant temperature of 55°C was measured, and the curve showed negligible fluctuation, showing that C4Q was stable (Fig. S17[†]). Furthermore, the cycling stability of the electrode materials is also an essential feature of rechargeable Al batteries. The Al/C4Q cell exhibits a high initial discharge capacity of 300 mA h g^{-1} and achieves considerable capacity retention (corresponding to a 0.111% capacity reduction of each cycle) at 0.1 A g^{-1} (Fig. S18[†]). Besides that, the cell exhibits a 261 mA h g^{-1} initial discharge capacity at 0.2 A g^{-1} and a 102 mA h g^{-1} retention over 500 cycles (Fig. 2d). Besides, C4Q likewise retains 87% and 76% after 1000 cycles at 0.5 and 1 A g^{-1} , respectively (Fig. S19 and S20[†]). SEM and TEM were performed on the pristine electrode, and the electrode was cycled with a negligible variation. Moreover, the element mappings show that the C and O elements in C4Q remain uniformly distributed, demonstrating that the morphology and structure of C4Q were

negligibly altered (Fig. S21–S23[†]). Furthermore, FTIR and XPS were also applied for the characterization of the stability of C4Q (Fig. S24 and S25[†]). These results demonstrate that the C4Q cathode maintains stability after cycling. The strong performance might be attributed to C4Q's adaptive construction, which can keep the electrode stable.

Likewise, to study the electrochemical kinetics process of the C4Q electrodes, CV tests are carried out at different scan rates within a range from 0.2 to 1.0 mV s^{-1} (Fig. 3a). Peak current (i) and scan rates (v) have the following relationship:

$$i = av^b$$

where v is the rate of scanning, i is the peak current, and a and b are values that can be adjusted. The b value is the slope of $\log(v)$ and $\log(i)$. When the b value is 0.5 , it indicates that the ion diffusion process controls the electrochemistry, while a b value of 1.0 suggests that surface capacity controls the behavior. In particular, the calculated b values for peak 1 and peak 4 are 0.75 and 0.76 , respectively (Fig. 3b). Besides, the capacitive and diffusion contributions to the total capacity at various scan rates were computed using CV curves (Fig. 3c, S26[†]). With scan rates ranging from 0.2 to 0.8 mV s^{-1} , it is evident that the C4Q electrode's capacitive contribution increases from 49 to 73.19% . These results indicate that the kinetics of the C4Q electrode was mainly controlled by the capacitance behaviour.

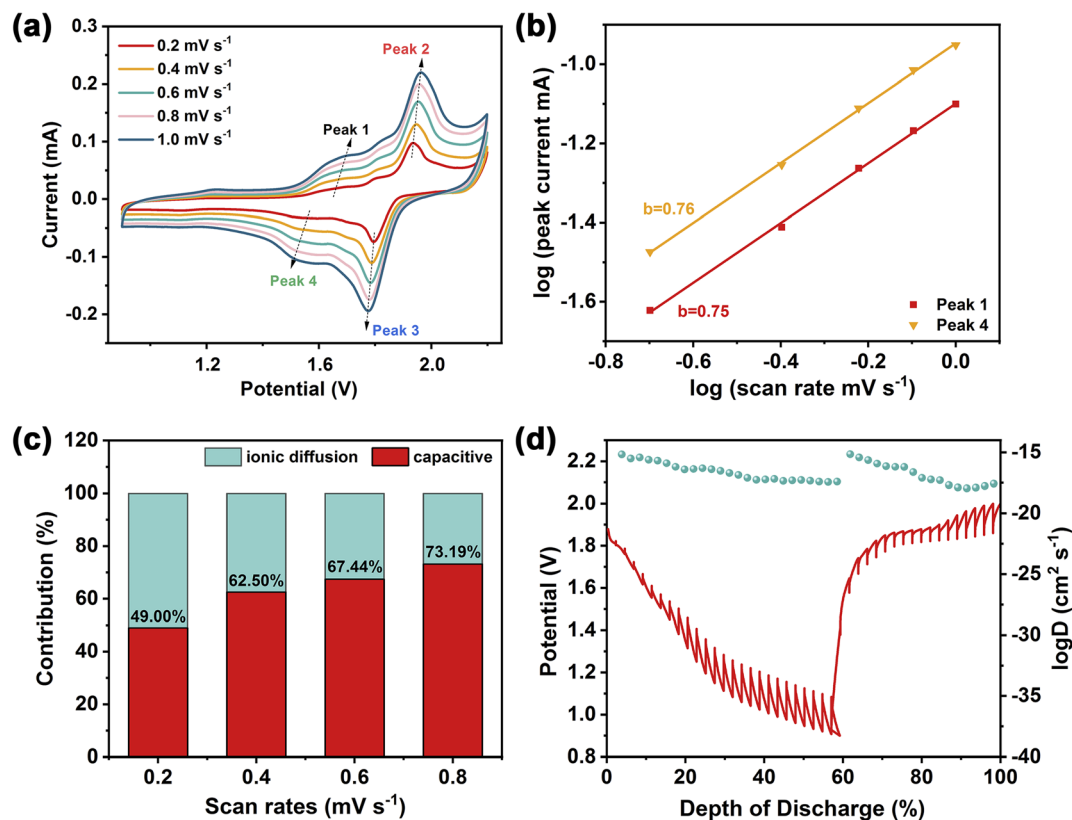


Fig. 3 (a) Different scan rates for CV profiles; (b) the matching plots of slope b of the redox peaks and $\log(\text{peak current}, i)$ vs. $\log(\text{scan rate}, v)$; (c) capacitive contributions in C4Q electrodes and the corresponding profiles at different scan rates; (d) galvanostatic intermittent titration technique (GITT) of the C4Q cathode.

Electrochemical impedance spectroscopy (EIS) was carried out, and the Nyquist plots are shown in Fig. S27.† In comparison with non-rotating materials, the results indicate less resistance and fast diffusion in term of electrochemical kinetics towards C4Q. To further illustrate the kinetic properties of active ion migration in the C4Q structure, the galvanostatic intermittent titration technique (GITT) test was carried out. For the GITT, the battery was allowed to rest for an hour after being discharged and then charged at a current of 50 mA g⁻¹ for 20 minutes. Specifically, the diffusion coefficients of active ions for the C4Q electrode are calculated based on the following equation according to previous work:⁴⁹

$$D = \frac{4}{\pi\tau} \left(\frac{v_m n_m}{S} \right)^2 \left(\frac{\Delta E_s}{\Delta E_t} \right)^2$$

here τ represents the pulse time, v_m represents the molar volume, n_m is the mole number, S is the area of the electrode interface, ΔE_s is the voltage change caused by the pulse, and ΔE_t is the change in the voltage of a galvanostatic current charge (discharge).⁴⁹ Fig. 3d exhibits the chemical diffusion coefficients obtained from the GITT test for the insertion and de-insertion of active ions. The results show that the diffusion coefficient of active ions from the C4Q electrode is 10⁻¹² cm² s⁻¹ (Fig. 3d), which is much superior to that of inorganic cathodes, such as Mo₆S₈ (10⁻¹⁹ cm² s⁻¹) and Co₃S₄ (10⁻²⁰ cm² s⁻¹).^{50,51}

Considering that organic liquid electrolytes are not conducive to the solubility of organic materials, a urea-based Lewis acid was used as the electrolyte in RABs. We conducted a ²⁷Al solid state NMR test for different concentrations of AlCl₃/urea electrolytes. The NMR spectra revealed that Al species included a urea-coordinated cation and chloroaluminate anions. The result of AlCl₃/urea = 1.0 (mole ratio) has obvious peaks located at 101.9 ppm, 88.6 ppm, and 72.4 ppm belonging to the AlCl₄⁻, [AlCl₃(urea)], and [AlCl₂(urea)₂]⁺ cations, respectively (Fig. S28a†). However, the spectrum (Fig. S28b†) of the AlCl₃/urea = 1.3 electrolyte shows a much broader peak than that of AlCl₃/urea = 1.0. The result of AlCl₃/urea = 1.3 might show the comprehensive presence of ions, including the AlCl₄⁻, [AlCl₃(urea)], and [AlCl₂(urea)₂]⁺ cations. Moreover, the intensity of the [AlCl₂(urea)₂]⁺ cation was more prominent, indicating that the concentration of the [AlCl₂(urea)₂]⁺ cation is higher and the active ion may be the [AlCl₂(urea)₂]⁺ cation in this electrolyte.

To study the Al-storage mechanism of the C4Q electrode during the charging and discharging process, the *ex situ* FTIR and *ex situ* XPS characterization studies are performed under different charge/discharge states (Fig. 4a). As shown in Fig. 4b, the *ex situ* FTIR spectra reveal that the peak located at 1680 cm⁻¹ belongs to the C=O bands, in which there was an obvious vibration peak of C=O at the initial state, and it gradually

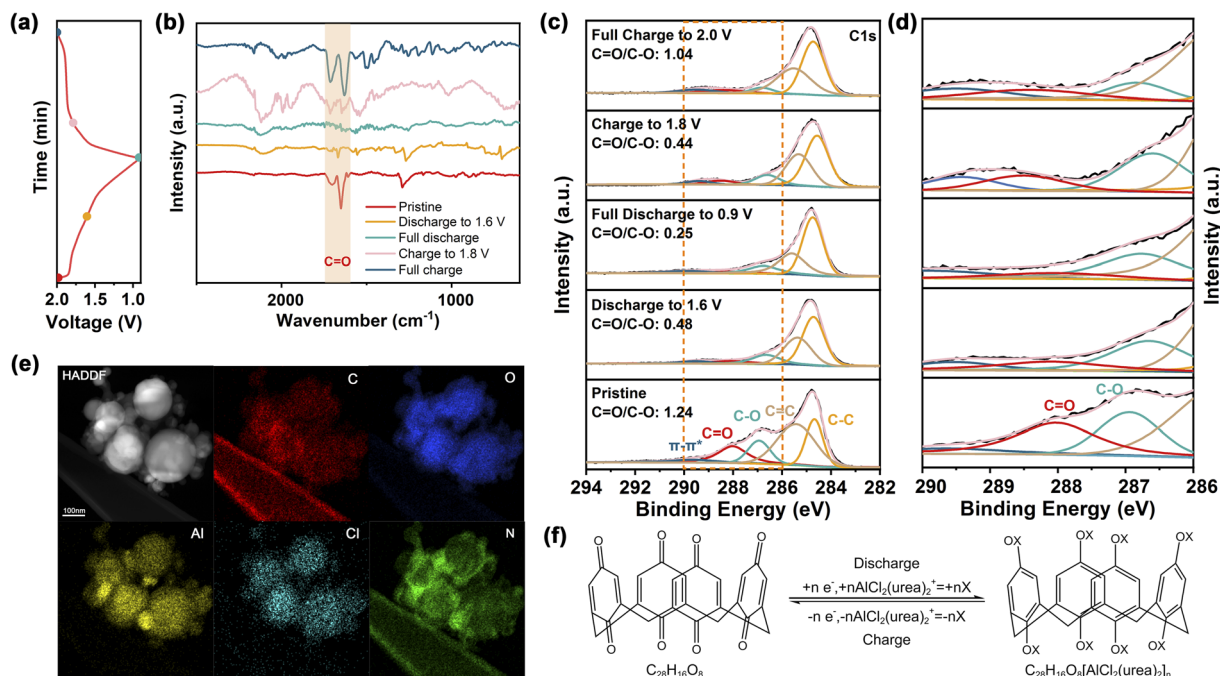


Fig. 4 Characterization of the reaction process. (a) Galvanostatic discharge/charge curve of the C4Q electrode; (b) *ex situ* ATR-FTIR spectra, (c) *ex situ* XPS profiles and (d) enlarged images of pristine, discharge to 1.8 V, full discharge, and charge to 1.6 V, and full charge of the C4Q cathode; high-angle annular dark field (HADDF) image and EDS color mapping images of C, O, N, Al, and Cl of the C4Q electrode in the (e) fully discharged state; (f) the redox mechanism of C4Q in an AlCl₃/urea electrolyte.

disappeared upon discharging and recovered during the charging process, indicating a good redox reaction reversibility of Al-based active ions ($[\text{AlCl}_2(\text{urea})_2]^+$) with C=O. These results demonstrate that the C=O band is the active site in C4Q. Moreover, *ex situ* XPS was measured to further study the reaction mechanism. The peaks of the C=O and C-O groups are situated at 288.1 eV and 287.0 eV, respectively, in the C 1s spectra, as shown in Fig. 4c. During the discharging stage, the ratio of C=O and C-O gradually decreases, while the ratio gradually increases in the charging state. When fully charged to 2.0 V, the ratio of C=O and C-O gradually recovered from the original 1.24 to 1.04. These results further verify that the active site of C4Q is C=O, and the reaction is reversible. The results are in accordance with the high-resolution XPS spectra of Al 2p, Cl 2p, and N 1s peaks (Fig. S29†). The Al 2p and Cl 2p peaks have a similar phenomenon; the intensity of the peaks increases during the process of discharging and decreases during charging. The residual signal was probably due to the electrolyte adsorption at the charge state. Moreover, the N 1s spectra show a similar phenomenon, and the peak increases with discharging and weakens with charging, which is similar to the changes of Al 2p and Cl 2p; that is to say, a reversible reaction is carried out in the process of charge and discharge. $[\text{AlCl}_2(\text{urea})_2]^+$ is shown as the working species in the elemental mappings of the C4Q electrode under the completely discharged condition (Fig. 4e). Under the complete discharge condition, regional enrichment of Al, Cl, and N elements is clearly visible, demonstrating the involvement of $[\text{AlCl}_2(\text{urea})_2]^+$ in the redox reaction during the discharge process. Forbye, the redox

mechanism of C4Q in AlCl₃/urea electrolyte as a cathode material for aluminium batteries is studied as shown in Fig. 4f. During discharge at the active site C=O on C4Q, the resultant electron is reduced to C-O and coupled with a positive active ion ($[\text{AlCl}_2(\text{urea})_2]^+$). And during charging, the C-O sites are oxidized to C=O, which are reversibly restored. This is the reason why the battery has a higher adsorption capacity of active ions and better stability in the battery performance. Consequently, *ex situ* FTIR, *ex situ* XPS, and *ex situ* TEM confirmed that C=O is the active site and the charging/discharging process is reversible.

Conclusions

In summary, to achieve high capacity and low steric resistance in RABs, a rotatable material (C4Q) is applied for the first time as cathode with sufficient utilization of active sites. Benefiting from the rotatable properties, the C4Q cathode obtained one of the highest energy densities of 480 W h kg⁻¹ (300 mA h g⁻¹) at 0.1 A g⁻¹. The process characterization studies demonstrated the active site in C4Q, which reversibly coordinated with $[\text{AlCl}_2(\text{urea})_2]^+$. The new rotatable organic materials with a low-cost electrolyte (AlCl₃/urea) provide a promising direction for RABs.

Author contributions

All authors contributed to the discussion of the content and edited the manuscript before submission.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

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